

# Assignment 2

October 16, 2024

## 0.1 Problem 1

### 0.2 1a)

```
[6]: import pandas as pd

df = pd.read_csv('shared/data/US_births_2000-2014.csv', parse_dates =_
    ↳ ['timestamp'])
print(df.head())
```

|   | year | month | day | day_of_week | births | timestamp  |
|---|------|-------|-----|-------------|--------|------------|
| 0 | 2000 | 1     | 1   | 6           | 9083   | 2000-01-01 |
| 1 | 2000 | 1     | 2   | 7           | 8006   | 2000-01-02 |
| 2 | 2000 | 1     | 3   | 1           | 11363  | 2000-01-03 |
| 3 | 2000 | 1     | 4   | 2           | 13032  | 2000-01-04 |
| 4 | 2000 | 1     | 5   | 3           | 12558  | 2000-01-05 |

1a) If births occur at the same rate all year round,  $1/7$  of birthdays should occur on any given day of the week. Approximately,  $1/12$  birthdays should occur in any given month.

### 0.3 1b)

```
[2]: #1b)
import matplotlib.pyplot as plt
import pandas as pd

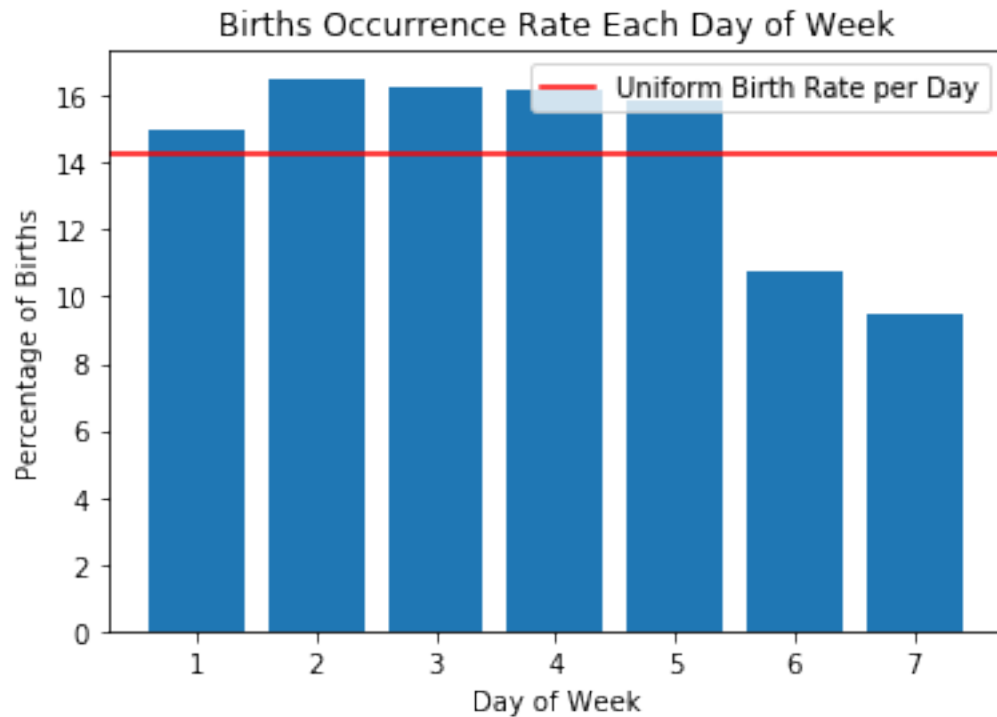
df = pd.read_csv('shared/data/US_births_2000-2014.csv', parse_dates =_
    ↳ ['timestamp'])

total_birthday_by_days = df.groupby('day_of_week')['births'].sum()

proportions = total_birthday_by_days/sum(total_birthday_by_days)

plt.bar(proportions.index,proportions*100)
plt.axhline(y=(1/7)*100,color='red',label='Uniform Birth Rate per Day')
plt.xlabel('Day of Week')
plt.ylabel('Percentage of Births')
plt.title('Births Occurrence Rate Each Day of Week')
plt.legend()
```

```
plt.show()
```



1b) The empirical proportions does not match the theoretical line. Birth rate on Saturday and Sunday below the theoretical rate, while birth rate on week day are higher than the theoretical rate. One possible explanation for such deviation from the theoretical rate could be that most doctor would not schedule a c-section surgery during the weekend, which artificially create this imbalance between the birth rate on weekdays and birth rate on weekends.

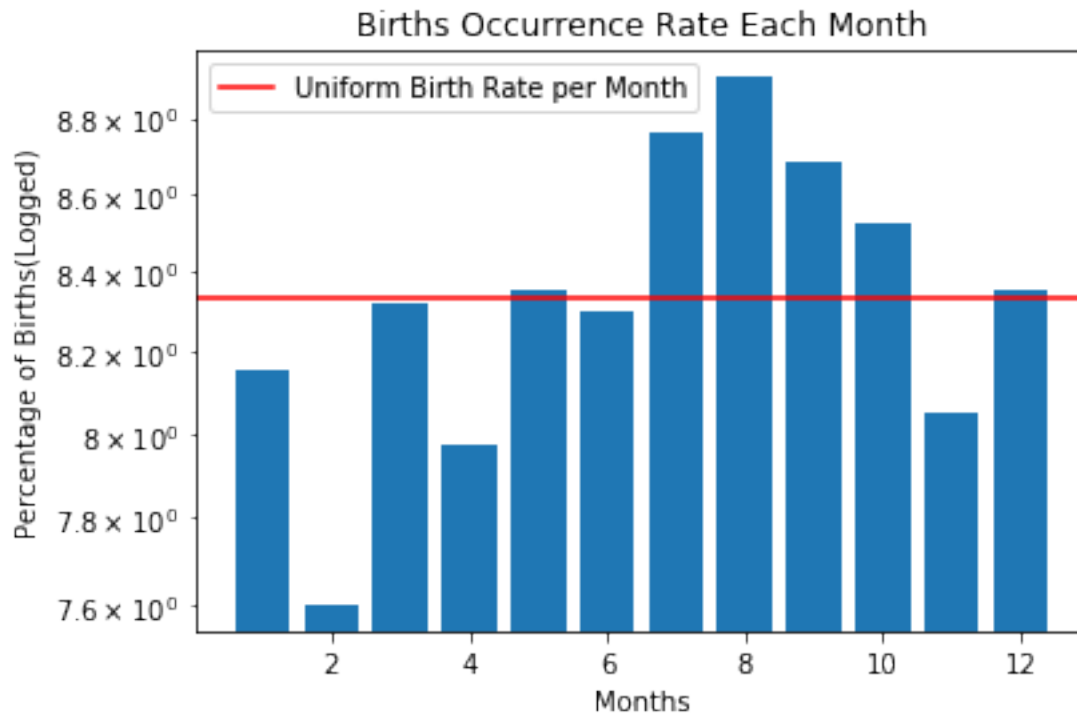
#### 0.4 1c)

```
[9]: #1c
import math

total_birthday_by_months = df.groupby('month')['births'].sum()
proportions = total_birthday_by_months/sum(total_birthday_by_days)

plt.bar(proportions.index,100*proportions)
plt.yscale('log')
plt.axhline(y=(100*(1/12)),color='red',label='Uniform Birth Rate per Month')
plt.xlabel('Months')
plt.ylabel('Percentage of Births(Logged)')
plt.title('Births Occurrence Rate Each Month')
plt.legend()
```

```
plt.show()
```

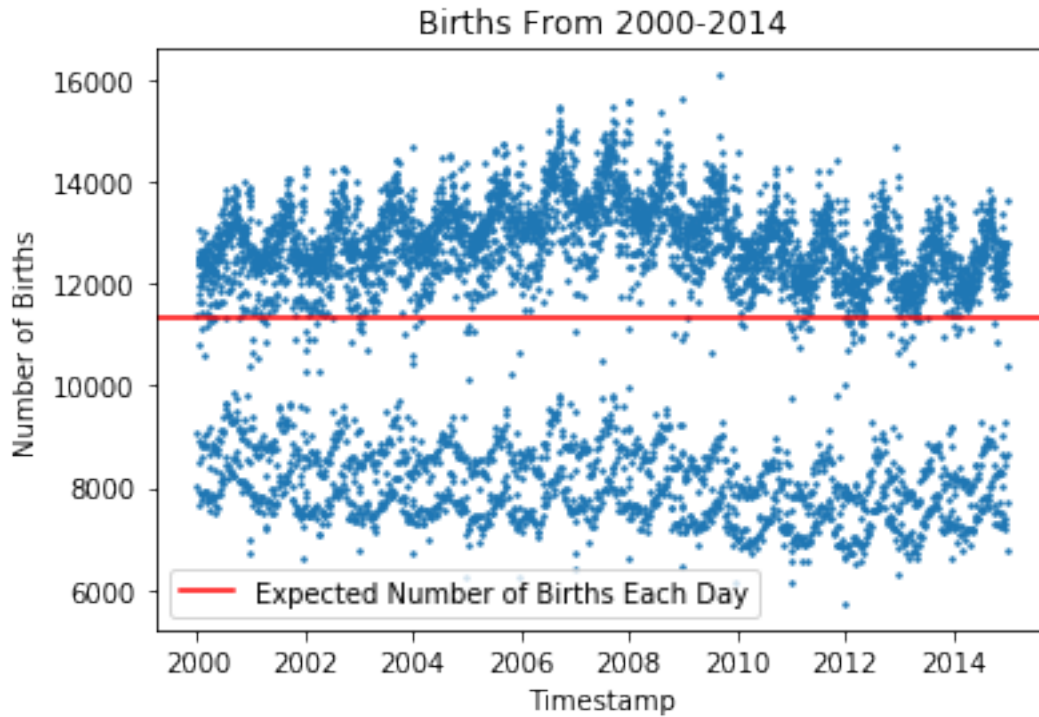


1c) The empirical proportions do not match the theoretical line for two reasons. First, the y-scale is logged, which tends to really show the differences between values when the values are relatively close to each other. One potential reason why February is so far off from other months is that February only has 28-29 days, while other months have 30-31 days. If the birth rate each month is entirely dependent on the days in a month, then January, which has 31 days, should be above the theoretical line! This means that another potential explanation for the differences in data could also be related to humans' mating patterns in different months.

## 0.5 1d)

```
[45]: #1d
plt.scatter(df.timestamp, df.births, s=2)
print(sum(df.births)/len(df))
plt.axhline(y=sum(df.births)/len(df), color='red', label='Expected Number of Births Each Day')
plt.xlabel('Timestamp')
plt.ylabel('Number of Births')
plt.title('Births From 2000-2014')
plt.legend()
plt.show()
```

11350.068260631502



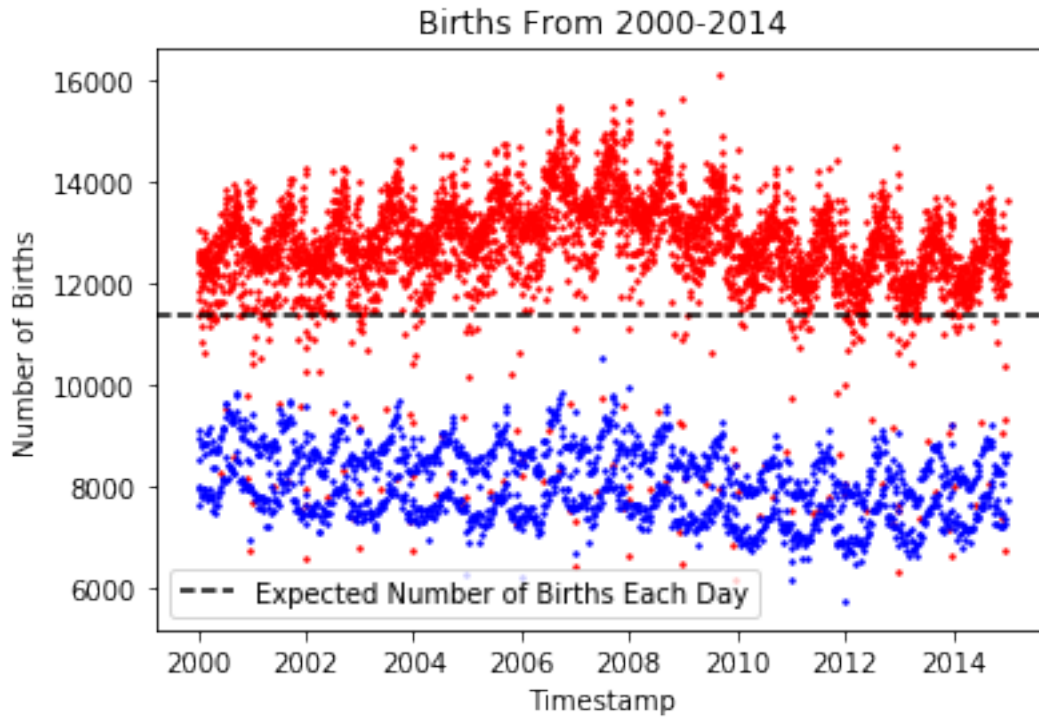
## 0.6 1e)

I think the lower group corresponds to the number of births on weekends.

```
[66]: #1e
births_M_to_F = df[df['day_of_week'].between(1, 5)][['births','timestamp']]
births_S_to_S = df[df['day_of_week'].between(6, 7)][['births','timestamp']]

plt.scatter(births_M_to_F.timestamp,births_M_to_F.births,s=2,color='red')
plt.scatter(births_S_to_S.timestamp,births_S_to_S.births,s=2,color='blue')

plt.axhline(y=sum(df.births)/len(df),
            color='black',linestyle='--',label='Expected Number of Births Each Day')
plt.xlabel('Timestamp')
plt.ylabel('Number of Births')
plt.title('Births From 2000-2014')
plt.legend()
plt.show()
```

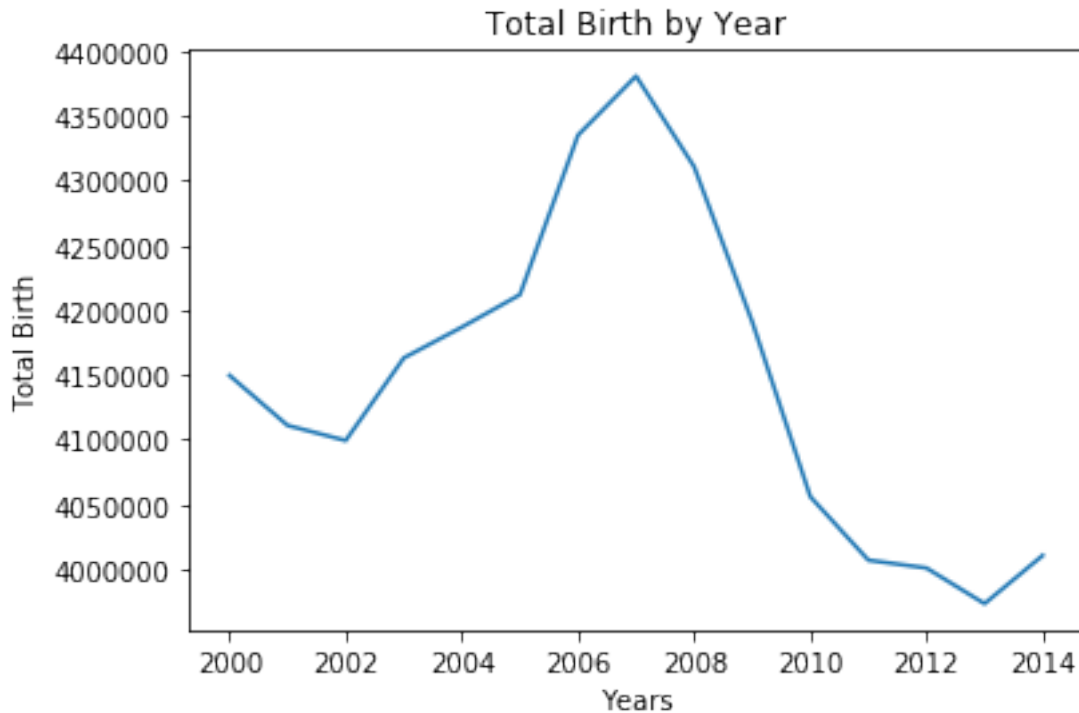


## 0.7 1f)

```
[14]: total_birth_by_year = df.groupby('year')['births'].sum()
      print(total_birth_by_year)
      plt.plot(total_birth_by_year.index, total_birth_by_year)
      plt.xlabel("Years")
      plt.ylabel("Total Birth")
      plt.title('Total Birth by Year')
      plt.show()
```

| year |         |
|------|---------|
| 2000 | 4149598 |
| 2001 | 4110963 |
| 2002 | 4099313 |
| 2003 | 4163060 |
| 2004 | 4186863 |
| 2005 | 4211941 |
| 2006 | 4335154 |
| 2007 | 4380784 |
| 2008 | 4310737 |
| 2009 | 4190991 |
| 2010 | 4055975 |
| 2011 | 4006908 |

```
2012    4000868
2013    3973337
2014    4010532
Name: births, dtype: int64
```



From the previous questions we could conclude that on a weekly scale, the birth rate tends to be higher on the weekdays and lower on the weekends. On a monthly scale, birth rate reach its height in August and its low in February. On a yearly scale, I chose a line graph to depict the trend in total births across 2000-2014. The total births/birth rates increase steadily from 2002-2007, reach its highest on 2007 and starts to decrease from 2008-2013. One potential explanation on the decrease birth rate from 2007 could be the famous 2008 financial crisis. Many are unemployed. Stock market performed terribly. Having a child during financially unstable time could be very stressful, hence the decrease in birth rate.

## 0.8 Problem 2

### 0.9 2a)

```
[2]: #2a
import pandas as pd

df=pd.read_csv('shared/data/framingham.csv')
df=df.dropna(0,subset=['cigsPerDay'])
print(df.head())
```

|   | male | age | education | currentSmoker | cigsPerDay | BPMeds | prevalentStroke | \ |
|---|------|-----|-----------|---------------|------------|--------|-----------------|---|
| 0 | 1    | 39  | 4.0       | 0             | 0.0        | 0.0    | 0               |   |
| 1 | 0    | 46  | 2.0       | 0             | 0.0        | 0.0    | 0               |   |
| 2 | 1    | 48  | 1.0       | 1             | 20.0       | 0.0    | 0               |   |
| 3 | 0    | 61  | 3.0       | 1             | 30.0       | 0.0    | 0               |   |
| 4 | 0    | 46  | 3.0       | 1             | 23.0       | 0.0    | 0               |   |

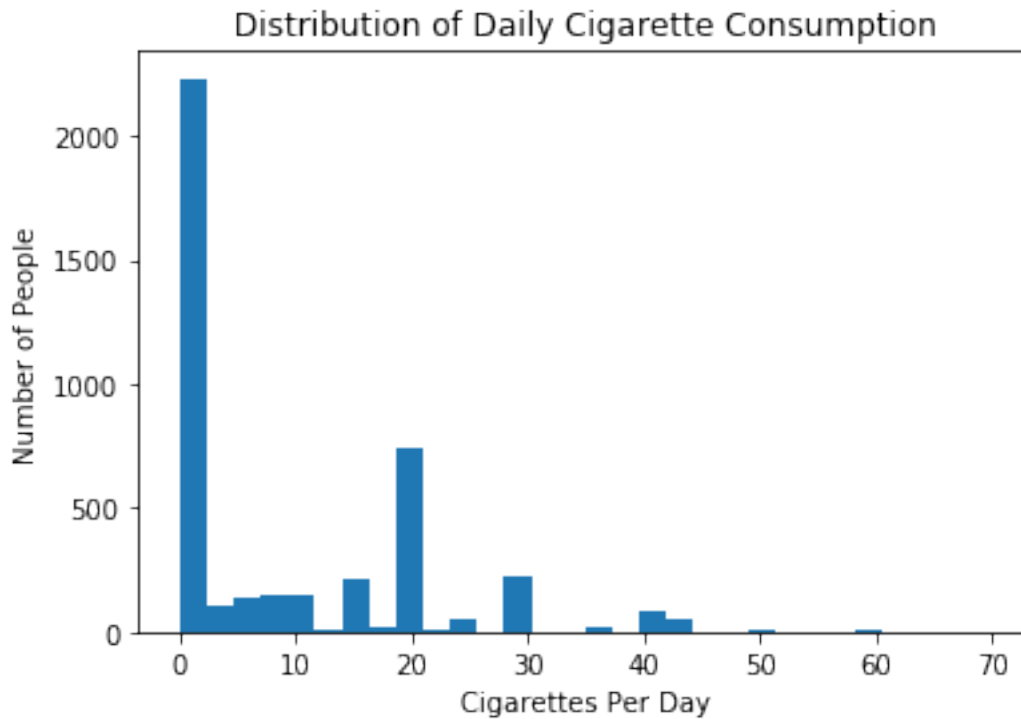
|   | prevalentHyp | diabetes | totChol | sysBP | diaBP | BMI   | heartRate | glucose | \ |
|---|--------------|----------|---------|-------|-------|-------|-----------|---------|---|
| 0 | 0            | 0        | 195.0   | 106.0 | 70.0  | 26.97 | 80.0      | 77.0    |   |
| 1 | 0            | 0        | 250.0   | 121.0 | 81.0  | 28.73 | 95.0      | 76.0    |   |
| 2 | 0            | 0        | 245.0   | 127.5 | 80.0  | 25.34 | 75.0      | 70.0    |   |
| 3 | 1            | 0        | 225.0   | 150.0 | 95.0  | 28.58 | 65.0      | 103.0   |   |
| 4 | 0            | 0        | 285.0   | 130.0 | 84.0  | 23.10 | 85.0      | 85.0    |   |

|   | TenYearCHD |
|---|------------|
| 0 | 0          |
| 1 | 0          |
| 2 | 0          |
| 3 | 1          |
| 4 | 0          |

## 0.10 2b)

```
[6]: #2b
import matplotlib.pyplot as plt

hist_2 = plt.hist(df.cigsPerDay,bins=30)
plt.xlabel("Cigarettes Per Day")
plt.ylabel("Number of People")
plt.title('Distribution of Daily Cigarette Consumption')
plt.show()
```



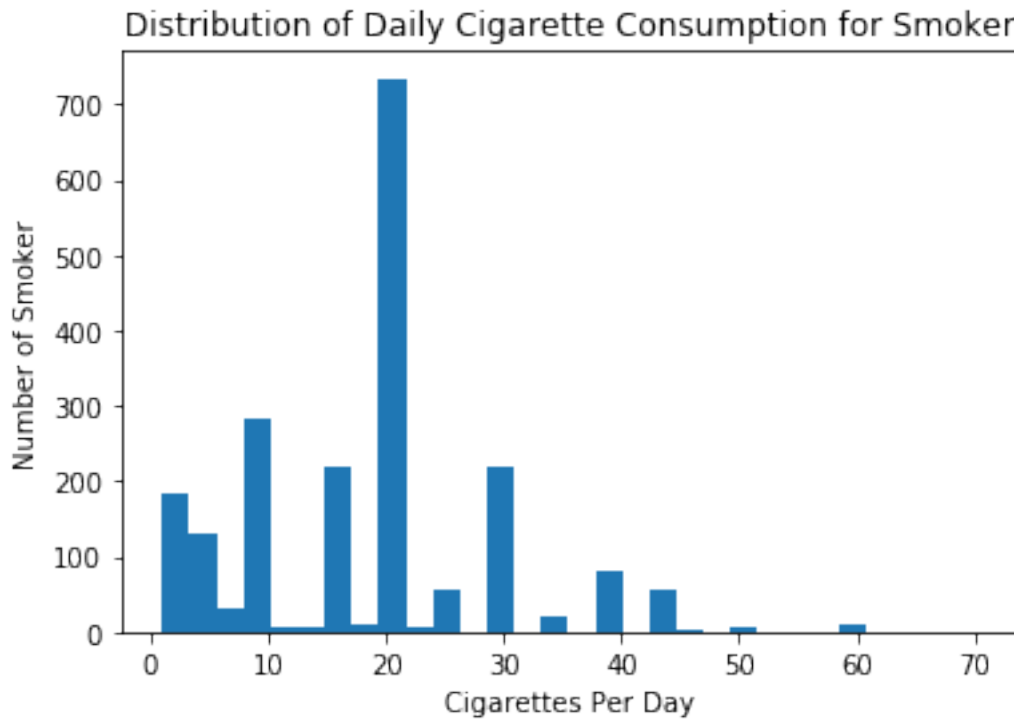
The histogram is skewed to the left, which means out of the sample population the majority of the sample tends to smoke less. There are few individuals who consume around 50-60 cigarettes per day, causing the graph to have a skinny long tail, which have huge effect on average and standard deviation.

## 0.11 2c)

```
[8]: #2c
smoker_freq = df[df.currentSmoker == 1]['cigsPerDay']

plt.hist(smoker_freq, bins = 30)
plt.xlabel("Cigarettes Per Day")
plt.ylabel("Number of Smoker")
plt.title('Distribution of Daily Cigarette Consumption for Smoker')
```

```
[8]: Text(0.5, 1.0, 'Distribution of Daily Cigarette Consumption for Smoker')
```



The histogram is skewed slightly to the left, but the distribution starts to resemble the normal distribution, while the previous graph has a severely skewed distribution. The median and average shifted towards the right, and standard deviation becomes smaller.

## 0.12 2d)

```
[9]: import numpy as np

print('Number of observations =',len(df.cigsPerDay))
print('The mean for cigsPerDay column =',round(np.mean(df.cigsPerDay)))
print('The median for cigsPerDay column =',np.median(df.cigsPerDay))
print('The standard deviation for cigsPerDay column =',round(np.std(df.
    ↪cigsPerDay)))
print('The max for cigsPerDay column =',np.max(df.cigsPerDay))
```

```
Number of observations = 4211
The mean for cigsPerDay column = 9.0
The median for cigsPerDay column = 0.0
The standard deviation for cigsPerDay column = 12.0
The max for cigsPerDay column = 70.0
```

### 0.13 2e)

As we have shown in the question above, the max for `cigsPerDay` is 70, which makes 300 an outlier. The mean will shift toward the right, while the median remains unchanged. The standard deviation will also increase drastically.

### 0.14 2f)

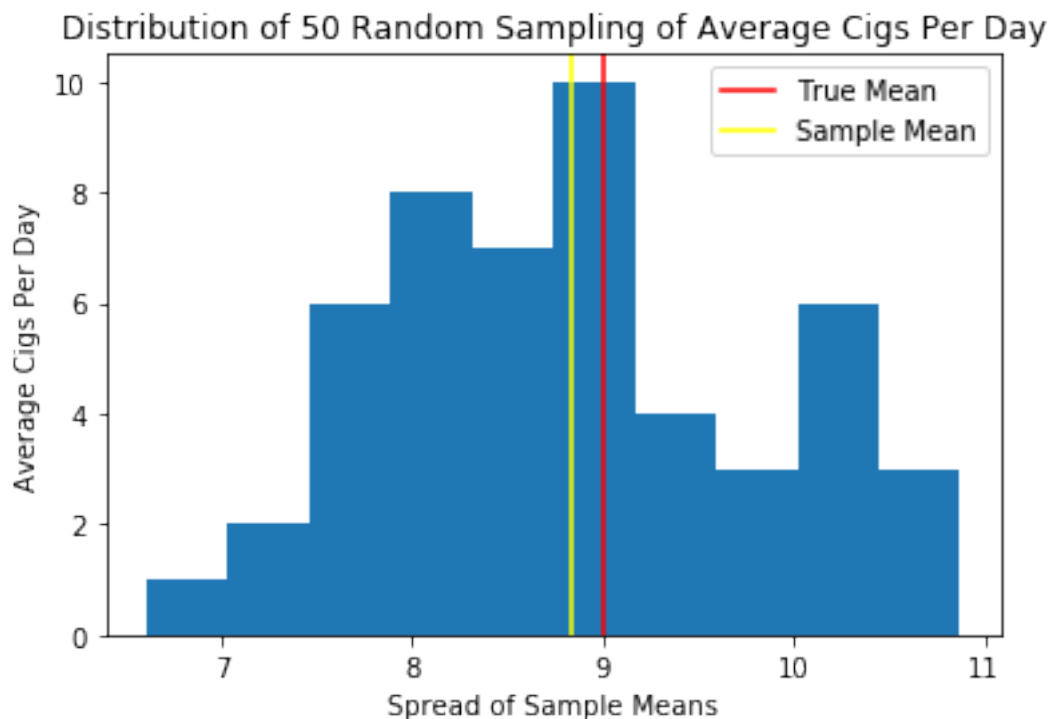
```
[30]: np.mean(np.random.choice(df.cigsPerDay,100))
```

```
[30]: 7.92
```

### 0.15 2g)

```
[14]: mean_list = []
      for i in range(50):
          mean_list.append(np.mean(np.random.choice(df.cigsPerDay,100)))

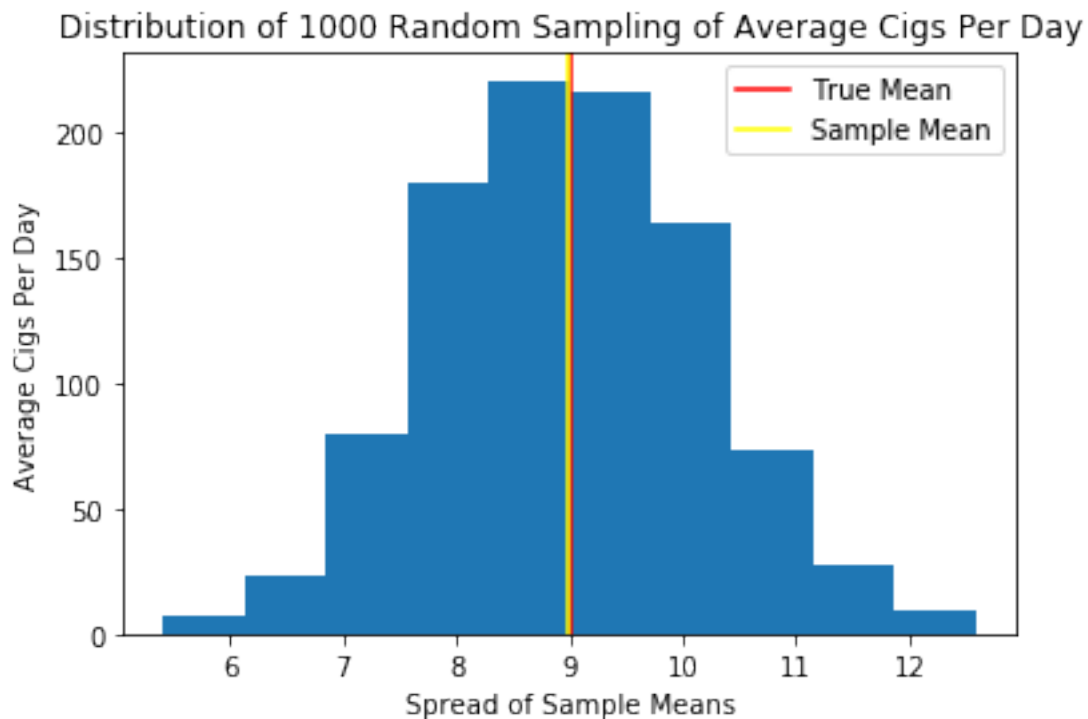
      plt.hist(mean_list)
      plt.axvline(x=9.0,label='True Mean',color='red')
      plt.axvline(x=sum(mean_list)/50,label='Sample Mean',color='Yellow')
      plt.xlabel('Spread of Sample Means')
      plt.ylabel('Average Cigs Per Day')
      plt.title('Distribution of 50 Random Sampling of Average Cigs Per Day')
      plt.legend()
      plt.show()
```



## 0.16 2h)

```
[38]: mean_list = []
      for i in range(1000):
          mean_list.append(np.mean(np.random.choice(df.cigsPerDay,100)))

      plt.hist(mean_list)
      plt.axvline(x=9.0,label='True Mean',color='red')
      plt.axvline(x=sum(mean_list)/1000,label='Sample Mean',color='Yellow')
      plt.xlabel('Spread of Sample Means')
      plt.ylabel('Average Cigs Per Day')
      plt.title('Distribution of 1000 Random Sampling of Average Cigs Per Day')
      plt.legend()
      plt.show()
```



## 0.17 2i)

The sample mean converges with the true population mean as the sample increases.

## 0.18 2j)

The shape of the sampling distribution becomes more symmetric and converges to the shape of normal distribution.

## 0.19 2k)

Law of large number and central limit theorem.

## 0.20 Problem 3

## 0.21 3a)

```
[30]: import pandas as pd
```

```
df = pd.read_csv('shared/data/federalist-papers/all_word_counts.csv')
print(df.head())
print(df.columns[0:6])
print(df.columns[len(df.columns)-10:len(df.columns)-5])
```

|   | from | the | more | general | inquiries | pursued | in | four | last | papers | ... | \ |
|---|------|-----|------|---------|-----------|---------|----|------|------|--------|-----|---|
| 0 | 12   | 184 | 7.0  | 3.0     | 2.0       | 1.0     | 33 | 1.0  | 2.0  | 1.0    | ... |   |
| 1 | 5    | 181 | 3.0  | 4.0     | 0.0       | 0.0     | 36 | 0.0  | 3.0  | 1.0    | ... |   |
| 2 | 13   | 202 | 6.0  | 1.0     | 0.0       | 0.0     | 65 | 0.0  | 0.0  | 0.0    | ... |   |
| 3 | 16   | 182 | 3.0  | 2.0     | 0.0       | 0.0     | 48 | 1.0  | 1.0  | 0.0    | ... |   |
| 4 | 10   | 211 | 13.0 | 4.0     | 0.0       | 1.0     | 58 | 1.0  | 0.0  | 0.0    | ... |   |

|   | scepticism | quite | insuring | aside | scales | Essay    | Author   | \ |
|---|------------|-------|----------|-------|--------|----------|----------|---|
| 0 | 0.0        | 0.0   | 0.0      | 0.0   | 0.0    | Essay 52 | Unknown  |   |
| 1 | 0.0        | 0.0   | 0.0      | 0.0   | 0.0    | Essay 67 | Hamilton |   |
| 2 | 0.0        | 0.0   | 0.0      | 0.0   | 0.0    | Essay 54 | Unknown  |   |
| 3 | 0.0        | 0.0   | 0.0      | 0.0   | 0.0    | Essay 21 | Hamilton |   |
| 4 | 0.0        | 0.0   | 0.0      | 0.0   | 0.0    | Essay 58 | Madison  |   |

|   | Publication         | Date       | total_word_count |
|---|---------------------|------------|------------------|
| 0 | New York Packet     | 1788-02-08 | 1841.0           |
| 1 | New York Packet     | 1788-03-11 | 1637.0           |
| 2 | New York Packet     | 1788-02-12 | 1994.0           |
| 3 | Independent Journal | NaN        | 1992.0           |
| 4 | Unknown             | NaN        | 2083.0           |

```
[5 rows x 8458 columns]
```

```
Index(['from', 'the', 'more', 'general', 'inquiries', 'pursued'],
      dtype='object')
```

```
Index(['scepticism', 'quite', 'insuring', 'aside', 'scales'], dtype='object')
```

### 0.22 3b)

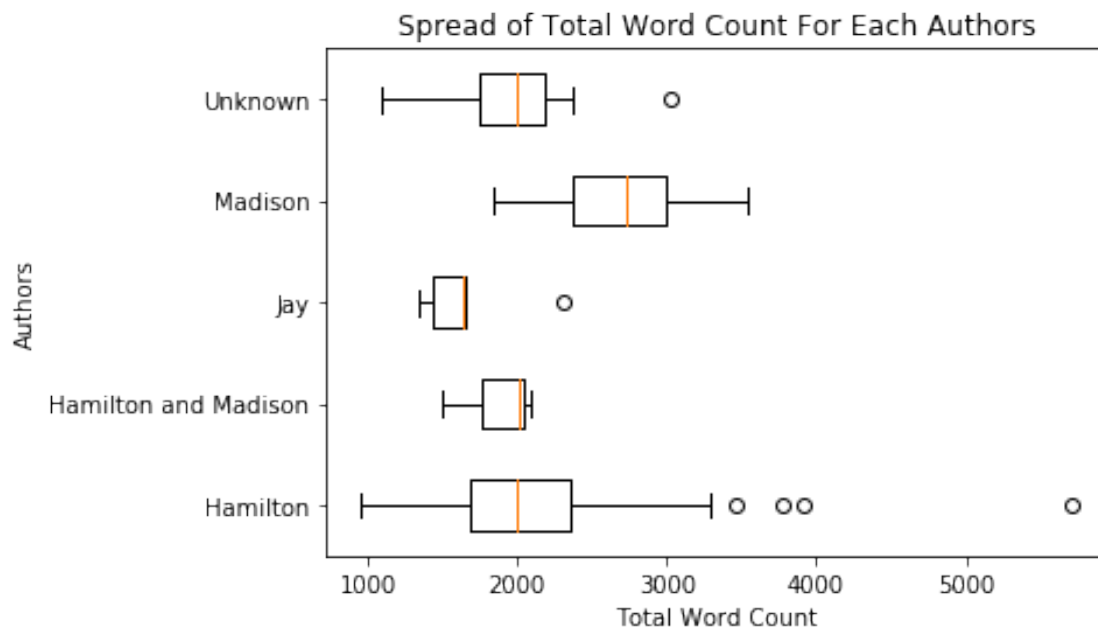
```
[31]: import numpy as np

Author = np.unique(df['Author'])
total = []
for author in Author:
    wordCount = df.loc[df['Author'] == author, 'total_word_count'].values
    total.append(wordCount)
```

First, I used NumPy's unique function to create a straightforward list of unique authors. To obtain a list of lists containing the total word count of each essay corresponding to each author, I initially tried the groupby() method. However, it only returned the sum of word counts for all essays by each author, which wasn't suitable for creating a box plot. Next, I attempted to use groupby() with Author and Essay set as indices, which provided the unique authors and the total word count for each essay, but I had trouble iterating through the data. After further research, I used the DataFrame's loc function, which allowed me to select the entire Author column. I then used a for loop to iterate through the list of authors, ensuring the sequence matched the original list. With df.loc, I filtered the rows to find those with the same author name, retrieving a list of total word counts for each essay by that author. Finally, I appended these word count lists to form a complete list of lists aligned with the author sequence.

### 0.23 3c)

```
[32]: plt.boxplot(total, labels = Author, vert =False)
plt.xlabel('Total Word Count')
plt.ylabel('Authors')
plt.title('Spread of Total Word Count For Each Authors')
plt.show()
```



I find it interesting that the median of unknown is similar to the median of Hamilton and Madison and Hamilton, even though the ranges for Hamilton is a lot larger than both. When Hamilton and Madison compose an essay together the distribution is skewed right heavily, and relatively shorter compared to their normal range.

## 0.24 3d)

```
[71]: ##i
grouped = df.groupby('Author')
df_Hamilton = grouped.get_group('Hamilton')

##ii
df_Hamilton_clean = df_Hamilton.drop(["Essay", 'Author', 'Publication', 'Date',
↳ 'total_word_count'], axis=1)

##iii
total=0
for i in df_Hamilton_clean.sum():
    total= total + i
print(total)
df_frequency = df_Hamilton_clean.sum()/total
print(df_frequency)

##iv
textGenerated=np.random.choice(df_frequency.index,size=250,p=df_frequency)

##v
joined = " ".join(textGenerated)
print(joined)
```

```
111035.0
from          0.005575
the           0.091980
more          0.002765
general       0.001234
inquiries     0.000099
...
scepticism    0.000009
quite         0.000009
insuring      0.000009
aside         0.000009
scales        0.000009
Length: 8453, dtype: float64
community would it union one places to subsequent the of incumbent to favor
extent incurred liberty to the fact the lights armies of beyond was a assign
```

that with in and it spirit the the navigation the state with up in men their  
yield this the will all course exportation mere day a cases to that republic  
states confederacy opposition conjunctures impracticability that we and a  
punishment can these improper of have purpose nation the information new far  
with against the object the that doubt sentence party governor rules from class  
of discussed other in which form times in in are those tempt no the removing be  
more be to impediments inspire immediate the other the duration the energy  
thereof conclude to connected and pretext of be some subject the local same the  
of are of certain revolution reading lending we requisites the a disposition  
feared such that taxes would first have an government in such establishments  
hand men of judicial must league of resentment of or the the the country our of  
that under that application ought single each power no it to theirs forth the  
national natural a governments persecution relate throughout for the that infer  
tincture tyrannical as so that in is of prohibitory states circumstance of  
assented and government for in revenue and or either been a included to a  
indicated will the it with class approved in of they but either other can with  
by exceptions to and in others particular who virginia year it

The result is not a realistic essay. First of all, punctuation is not accounted in the passage, which breaks the logic inherit in the structure of language. In order for it to sound like a realistic essay, the structure of language like grammars and punctuation needs to be followed. Second, the author did not account the contextual word, which can be crucial for a natural passage. The contextual words that appear frequently is the main theme or point that the author is trying to convey. Without such words the passage seems unnatural because a huge part of language is to convey some sort of meaning.

## 0.25 Problem 4

### 0.26 4a)

It means that when one of the winner is chosen, his/her tickets will be put back in the raffles and have the possibility to win again. The probability of one person winning all \$300 is  $1/100 * 1/100 * 1/100$ , which is  $1/1000000$ .

### 0.27 4b)

If the lottery draws with replacement, then the probability of winning at least once is  $3/100$ .

### 0.28 4c)

The total group will be one smaller when one winner is drawn out. If the lottery draws without replacement, then the probability of winning at least one is  $1/100 + 1/99 + 1/98$ .

### 0.29 4d)

```
[18]: print('The probability of all three of us win is',round(100*((1/100)*(1/99)*(1/
↪98)),7),'percent')
```

The probability of all three of us win is 0.0001031 percent

0.30 4e)

```
[86]: print(3/100)
      print((1/100)+(1/99)+(1/98))
      print(100*((1/100)+(1/99)+(1/98)-3/100))
```

```
0.03
0.030305091733663164
0.030509173366316533
```

The difference between with replacement and without replacement in terms of probability is 0.030509173366316533 percent, with without replacement having a slightly greater chance to win. The difference mainly comes from the variation of the denominator.

```
[21]: n=10000000
      (((1/100) *(100-4))*n -4*n)/n
```

```
[21]: -3.04
```

According to the law of large numbers, the sample winning probability will converge into the expected winning probability when the sample is large enough, which means the probability of winning would be close to 1/100. If I enter every day, and the ticket costs \$4, then out of every hundred times, I would win 96 dollars. For every 400 dollars I spend, I would gain 96 dollars, which means every ticket has an average profit of -3.04 dollars.